Week 3: Independence, Random Variables, and Distributions

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Reading: Casella & Berger (C&B) §1.3 (independence, pp. 24–27) and §§1.4–1.6

Lecture	Date	Topics	C&B
3.1	Tue. Oct. 13	Independence of two and of several events; random variables; induced probability functions	§1.3 (end), §1.4
3.2	Thu. Oct. 15	Cumulative distribution functions; continuous and discrete random variables; identically distributed random variables; probability mass and density functions	§§1.5–1.6

Notation. Random variables are uppercase (X, Y); their realized values are lowercase (x, y). $\mathcal{X}$ is the range of X. F_X is the cdf of X and f_X its pmf or pdf. $\mathcal{B}^1$ is the Borel sigma algebra on the real line: the smallest sigma algebra containing all intervals. $X \sim F_X(x)$ reads "X is distributed as $F_X(x)$."

Independence

> **Way of thinking.** Translate intuition into equations.

1. Independence of two events

Sometimes learning B tells us nothing about A: $P(A \mid B) = P(A)$. By Bayes' rule (1.3.5) this implies $P(B \mid A) = P(B)$, and by the multiplication rule $P(A \cap B) = P(A)P(B)$. We take the symmetric version as the definition.

Definition 1.3.7. Two events, A and B, are **statistically independent** if

$$P(A \cap B) = P(A)\,P(B). \tag{1.3.8}$$

This definition is symmetric in A and B, does not require $P(B) > 0$, and extends to more than two events.

Example 1.3.8 (Chevalier de Méré). Is it a good bet that at least one six appears in 4 rolls of a die? By independence of the rolls,

$$P(\text{at least one 6 in 4 rolls}) = 1 - P(\text{no 6 in 4 rolls}) = 1 - \left(\frac{5}{6}\right)^4 \approx 0.518.$$

De Méré also bet on at least one double six in 24 rolls of two dice, reasoning that it should be equally likely. In fact $1 - (35/36)^{24} \approx 0.491$, a losing bet. His losses prompted the Pascal–Fermat correspondence of 1654, often taken as the birth of probability theory.

Theorem 1.3.9. If A and B are independent events, then the following pairs are also independent:

- **(a)** A and B^c,
- **(b)** A^c and B,
- **(c)** A^c and B^c.

Proof of (a). By Theorem 1.2.9(a) and independence,

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)\big(1 - P(B)\big) = P(A)P(B^c). \qquad \square$$

Parts (b) and (c) are Exercise 1.40.

Independent vs. disjoint. These are very different. If $P(A) > 0$ and $P(B) > 0$ and A, B are disjoint, then $P(A \cap B) = 0 \neq P(A)P(B)$, so they are **dependent**: learning that A occurred tells you B did not (Exercise 1.39).

2. Independence of several events

What should it mean for A, B, and C to be independent? Two natural guesses both fail.

Guess 1: $P(A \cap B \cap C) = P(A)P(B)P(C)$ is enough. False.

Example 1.3.10 (Tossing two dice). Roll two dice, with S the 36 equally likely ordered pairs. Let A = "doubles," B = "the sum is between 7 and 10," and C = "the sum is 2, 7, or 8." Then $P(A) = 1/6$, $P(B) = 1/2$, $P(C) = 1/3$, and

$$P(A \cap B \cap C) = P(\{(4,4)\}) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{3}.$$

But $P(B \cap C) = P(\text{sum equals 7 or 8}) = 11/36 \neq P(B)P(C) = 1/6$. Similarly, $P(A \cap B) \neq P(A)P(B)$, so the triple product condition does not even guarantee pairwise independence.

Guess 2: pairwise independence is enough. False.

Example 1.3.11 (Letters). Let S consist of the 3! permutations of the letters a, b, and c along with the three triples of each letter, each with probability $1/9$:

$$S = \{aaa,\ bbb,\ ccc,\ abc,\ bca,\ cba,\ acb,\ bac,\ cab\}.$$

Let A_i = "the ith place in the triple is occupied by a." Then $P(A_i) = 1/3$ and $P(A_i \cap A_j) = 1/9$ for all $i \neq j$, so the A_i are pairwise independent. But $P(A_1 \cap A_2 \cap A_3) = P(\{aaa\}) = 1/9 \neq 1/27$.

Definition 1.3.12. A collection of events $A_1, \ldots, A_n$ are **mutually independent** if for any subcollection $A_{i_1}, \ldots, A_{i_k}$, we have

$$P\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} P(A_{i_j}).$$

For three events that is four equations: three pairwise conditions plus the triple one. For n events it is $2^n - n - 1$ equations.

Example 1.3.13 (Three coin tosses–I). Toss a coin three times, with sample space

$$S = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\},$$

and assign probability $1/8$ to each point. Let H_i = "toss i is a head," for example $H_1 = \{HHH, HHT, HTH, HTT\}$. Then $P(H_i) = 1/2$,

$$P(H_1 \cap H_2 \cap H_3) = P(\{HHH\}) = \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2},$$

$$P(H_1 \cap H_2) = P(\{HHH, HHT\}) = \frac{2}{8} = \frac{1}{2} \cdot \frac{1}{2},$$

and similarly for the other pairs. So H_1, H_2, H_3 are mutually independent: a head on any toss has no effect on any other toss. Conversely, assigning $1/8$ to each point is the *only* probability model with $P(H_i) = 1/2$ and H_1, H_2, H_3 mutually independent. This is what "three independent tosses of a fair coin" means.

Check your understanding.

1. If $P(A) = 0.4$, $P(B) = 0.5$, and A, B are independent, find $P(A \cup B)$.

   > $P(A \cup B) = P(A) + P(B) - P(A)P(B) = 0.4 + 0.5 - 0.2 = 0.7$. Alternatively, by Theorem 1.3.9(c), $P(A \cup B) = 1 - P(A^c)P(B^c) = 1 - 0.6 \times 0.5 = 0.7$.

2. Show that if $P(B) = 1$, then B is independent of every event A.

   > $P(A \cap B^c) \leq P(B^c) = 0$, so $P(A \cap B) = P(A) - P(A \cap B^c) = P(A) = P(A)P(B)$.

3. A system has three components that work independently, each with probability 0.9. It works if at least one component works. What is the probability the system works?

   > By Theorem 1.3.9 the complements are also mutually independent, so $P(\text{all fail}) = 0.1^3 = 0.001$ and $P(\text{system works}) = 0.999$.

Random Variables

> **Way of thinking.** A random variable is neither random nor a variable: it is a *function* on the sample space.

3. Definition and examples

In many experiments it is easier to work with a summary than with the original sample space. Ask 50 people whether they agree (1) or disagree (0) with a statement. The sample space has 2^{50} elements, ordered strings of 0s and 1s. If all we care about is X = the number of people who agree, the new sample space is just $\{0, 1, \ldots, 50\}$.

Definition 1.4.1. A **random variable** is a function from a sample space S into the real numbers.

Example 1.4.2 (Random variables). In some experiments random variables are implicitly used:

Experiment	Random variable
Toss two dice	X = sum of the numbers
Toss a coin 25 times	X = number of heads in 25 tosses
Apply different amounts of fertilizer to corn plants	X = yield/acre

4. The induced probability function

Defining a random variable also defines a new sample space, its range $\mathcal{X}$. We must check that the probability function P on S can be used for X.

Suppose $S = \{s_1, \ldots, s_n\}$ with probability function P, and X has range $\mathcal{X} = \{x_1, \ldots, x_m\}$. We observe $X = x_i$ if and only if the outcome is an $s_j \in S$ with $X(s_j) = x_i$. So we define

$$P_X(X = x_i) = P\big(\{s_j \in S : X(s_j) = x_i\}\big). \tag{1.4.1}$$

P_X is an **induced** probability function on $\mathcal{X}$, defined in terms of P. It satisfies the Kolmogorov Axioms (Exercise 1.45). Because of (1.4.1) we simply write $P(X = x_i)$ rather than $P_X(X = x_i)$.

A note on notation. Random variables are always denoted with uppercase letters and their realized values (or range) with the corresponding lowercase letters. The random variable X can take the value x.

Example 1.4.3 (Three coin tosses–II). Toss a fair coin three times (Example 1.3.13) and let X = number of heads.

s	HHH	HHT	HTH	THH	TTH	THT	HTT	TTT
$X(s)$	3	2	2	2	1	1	1	0

The range is $\mathcal{X} = \{0, 1, 2, 3\}$. With probability $1/8$ on each point, counting gives

x	0	1	2	3
$P_X(X = x)$	$1/8$	$3/8$	$3/8$	$1/8$

For example, $P_X(X = 1) = P(\{HTT, THT, TTH\}) = 3/8$.

Example 1.4.4 (Distribution of a random variable). We can find P_X even without a complete listing. Let S be the 2^{50} equally likely strings of 50 0s and 1s, X = number of 1s, and $\mathcal{X} = \{0, 1, \ldots, 50\}$. Then

$$P_X(X = 27) = \frac{\text{number of strings with 27 1s}}{\text{number of strings}} = \frac{\binom{50}{27}}{2^{50}},$$

and in general,

$$P_X(X = i) = \frac{\binom{50}{i}}{2^{50}}, \quad i \in \mathcal{X}.$$

This is Week 2's counting at work.

Uncountable range. If $\mathcal{X}$ is uncountable, we define P_X in the same way as (1.4.1): for any set $A \subset \mathcal{X}$,

$$P_X(X \in A) = P\big(\{s \in S : X(s) \in A\}\big). \tag{1.4.2}$$

This is a legitimate probability function. (To be precise, (1.4.2) defines probabilities only for a certain sigma algebra of subsets of $\mathcal{X}$, the Borel sets. C&B do not dwell on this, and neither will we.)

The key picture. $\{X \in A\}$ is shorthand for the *event* $\{s \in S : X(s) \in A\}$, the inverse image of A under X. Every probability statement about X is a statement about an event in S.

Check your understanding.

1. Roll two fair dice and let X = the sum. Write the event $\{X = 4\}$ as a subset of S and find $P(X = 4)$.

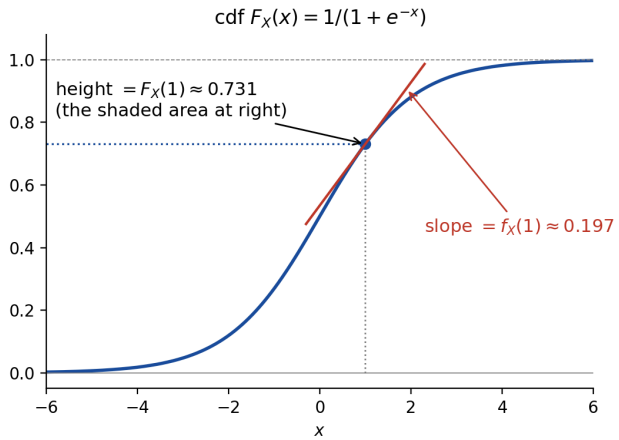
   > $\{X = 4\} = \{(1,3),(2,2),(3,1)\}$, so $P(X = 4) = 3/36 = 1/12$.

2. In Example 1.4.3, let $Y = X^2$. Is Y a random variable? Find the range of Y and $P(Y = 4)$.

   > Yes: $Y(s) = X(s)^2$ is again a function from S into the real numbers. Its range is $\{0, 1, 4, 9\}$ and $P(Y = 4) = P(X = 2) = 3/8$. Functions of random variables are the topic of Chapter 2.

Distribution Functions, Mass Functions, and Density Functions

> **Way of thinking.** cdf vs. pdf



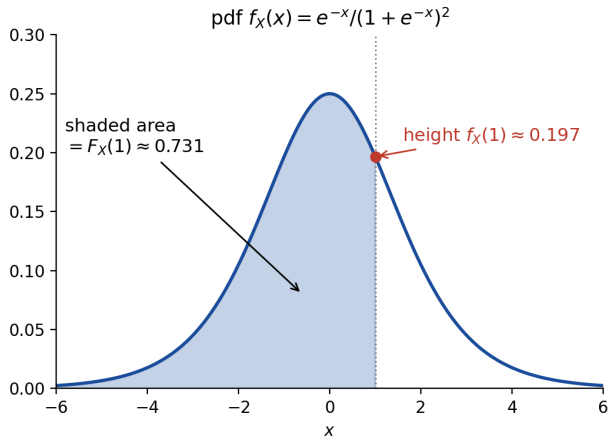


Figure. The cdf (left) and pdf (right) of the logistic distribution (Examples 1.5.5 and 1.6.4). The **height** of the cdf at x equals the **area** under the pdf to the left of x: $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$. Going back, the **slope** of the cdf at x equals the **height** of the pdf: $f_X(x) = \frac{d}{dx} F_X(x)$. At $x = 1$: $F_X(1) \approx 0.731$ and $f_X(1) \approx 0.197$.

5. The cumulative distribution function

Definition 1.5.1. The **cumulative distribution function** or **cdf** of a random variable X, denoted by $F_X(x)$, is defined by

$$F_X(x) = P_X(X \le x), \quad \text{for all } x.$$

Example 1.5.2 (Tossing three coins). Toss three fair coins and let X = number of heads. The cdf of X is

$$F_X(x) = \begin{cases} 0 & \text{if } -\infty < x < 0 \\ \frac{1}{8} & \text{if } 0 \le x < 1 \\ \frac{1}{2} & \text{if } 1 \le x < 2 \\ \frac{7}{8} & \text{if } 2 \le x < 3 \\ 1 & \text{if } 3 \le x < \infty. \end{cases} \tag{1.5.1}$$

- F_X is defined for **all** real x, not just $x \in \mathcal{X} = \{0, 1, 2, 3\}$. For example, $F_X(2.5) = P(X \le 2.5) = P(X = 0, 1, \text{ or } 2) = 7/8$.
- F_X is a step function. It jumps at each $x_i \in \mathcal{X}$, and the size of the jump at x_i is $P(X = x_i)$.
- $F_X(x) = 0$ for $x < 0$ since X cannot be negative, and $F_X(x) = 1$ for $x \ge 3$.

Right-continuity. At a jump point, F_X takes the value at the *top* of the jump (note the inequalities $\le$ and $<$ in (1.5.1)). So F_X is continuous when a point is approached from the right. This follows from the "$\le$" in the definition. Had we defined $F_X(x) = P_X(X < x)$, F_X would be left-continuous. In general, the size of the jump at any point x is

$$P(X = x) = F_X(x) - \lim_{t \uparrow x} F_X(t).$$

Theorem 1.5.3. The function $F(x)$ is a cdf if and only if the following three conditions hold:

- **a.** $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.
- **b.** $F(x)$ is a nondecreasing function of x.
- **c.** $F(x)$ is right-continuous; that is, for every number x_0, $\lim_{x \downarrow x_0} F(x) = F(x_0)$.

Outline of proof. *Necessity* (every cdf has these properties) follows by writing F in terms of the probability function (Exercise 1.48). For example, (b) holds because $x_1 < x_2$ implies $\{X \le x_1\} \subset \{X \le x_2\}$, so $F(x_1) \le F(x_2)$ by Theorem 1.2.9(c). Parts (a) and (c) need the continuity of probability along monotone sequences of events. *Sufficiency* (any F with these

properties is the cdf of some random variable) is much harder: we must construct a sample space S, a probability function P on S, and a random variable X on S whose cdf is F. □

Why this theorem matters. To specify a distribution, we do not need to build S and P explicitly. Any function with properties (a)–(c) will do.

Example 1.5.4 (Tossing for a head). Toss a coin until a head appears, where p = probability of a head on any given toss, and let X = number of tosses required to get a head. For $x = 1, 2, \ldots$,

$$P(X = x) = (1-p)^{x-1}p, \tag{1.5.2}$$

since we need $x-1$ tails followed by a head, and the tosses are independent. For any positive integer x,

$$P(X \le x) = \sum_{i=1}^{x} P(X = i) = \sum_{i=1}^{x} (1-p)^{i-1}p. \tag{1.5.3}$$

Using the partial sum of the geometric series (Exercise 1.50),

$$\sum_{k=1}^{n} t^{k-1} = \frac{1-t^n}{1-t}, \quad t \neq 1, \tag{1.5.4}$$

we get

$$F_X(x) = P(X \le x) = \frac{1-(1-p)^x}{1-(1-p)}\, p = 1-(1-p)^x, \quad x = 1, 2, \ldots.$$

Between the integers F_X is flat, so for all real $x \ge 1$, $F_X(x) = 1-(1-p)^{\lfloor x \rfloor}$, and $F_X(x) = 0$ for $x < 1$.

If $0 < p < 1$, F_X satisfies Theorem 1.5.3:

- **(a)** $F_X(x) = 0$ for all $x < 0$, and $\lim_{x\to\infty} 1-(1-p)^x = 1$.
- **(b)** The sum in (1.5.3) contains more positive terms as x increases.
- **(c)** For any x, $F_X(x+\epsilon) = F_X(x)$ if $\epsilon > 0$ is small enough, so $\lim_{\epsilon \downarrow 0} F_X(x+\epsilon) = F_X(x)$.

This is the cdf of the **geometric distribution** (named after the series).

Example 1.5.5 (Continuous cdf). The function

$$F_X(x) = \frac{1}{1+e^{-x}} \tag{1.5.5}$$

satisfies Theorem 1.5.3: $\lim_{x\to-\infty} F_X(x) = 0$ since $\lim_{x\to-\infty} e^{-x} = \infty$; $\lim_{x\to\infty} F_X(x) = 1$ since $\lim_{x\to\infty} e^{-x} = 0$; and

$$\frac{d}{dx}F_X(x) = \frac{e^{-x}}{(1+e^{-x})^2} > 0,$$

so F_X is increasing. It is not only right-continuous but continuous. This is a special case of the **logistic distribution**, the same function as the logistic (sigmoid) link in logistic regression.

Example 1.5.6 (Cdf with jumps). A cdf can mix continuous pieces and jumps. For some ϵ with $1 > \epsilon > 0$, let

$$F_Y(y) = \begin{cases} \dfrac{1-\epsilon}{1+e^{-y}} & \text{if } y < 0 \\ \\ \epsilon + \dfrac{1-\epsilon}{1+e^{-y}} & \text{if } y \geq 0. \end{cases} \tag{1.5.6}$$

F_Y has a jump of height ϵ at $y = 0$ and is otherwise continuous (Exercise 1.47). Think of a gauge whose reading could be any real number but that sometimes sticks at 0, with ϵ the probability that it sticks.

6. Continuous and discrete random variables

Definition 1.5.7. A random variable X is **continuous** if $F_X(x)$ is a continuous function of x. A random variable X is **discrete** if $F_X(x)$ is a step function of x.

So Examples 1.5.2 and 1.5.4 are discrete, Example 1.5.5 is continuous, and Example 1.5.6 is neither.

Definition 1.5.8. The random variables X and Y are **identically distributed** if, for every set $A \in \mathcal{B}^1$, $P(X \in A) = P(Y \in A)$.

Identically distributed does **not** mean equal: Definition 1.5.8 does not say $X = Y$.

Example 1.5.9 (Identically distributed random variables). Toss a fair coin three times, and let X = number of heads and Y = number of tails. For each $k = 0, 1, 2, 3$, $P(X = k) = P(Y = k)$, so X and Y are identically distributed. However, $X(s) \neq Y(s)$ for every sample point s, since $X(s) + Y(s) = 3$ is odd.

Theorem 1.5.10. The following two statements are equivalent:

- **a.** The random variables X and Y are identically distributed.
- **b.** $F_X(x) = F_Y(x)$ for every x.

Proof. (a) $\Rightarrow$ (b): For every x, the set $(-\infty, x]$ is in $\mathcal{B}^1$, so

$$F_X(x) = P\big(X \in (-\infty, x]\big) = P\big(Y \in (-\infty, x]\big) = F_Y(x).$$

b. $\Rightarrow$ (a) is much harder: agreement on all intervals $(-\infty, x]$ must be extended to agreement on all sets in $\mathcal{B}^1$. This needs heavy use of sigma algebras, and we omit it (Chung 1974, Section

2.2). □

The takeaway: the cdf completely determines the distribution of a random variable.

7. Probability mass functions

Definition 1.6.1. The **probability mass function** (**pmf**) of a discrete random variable X is given by

$$f_X(x) = P(X = x) \quad \text{for all } x.$$

Example 1.6.2 (Geometric probabilities). For the geometric distribution of Example 1.5.4,

$$f_X(x) = P(X = x) = \begin{cases} (1-p)^{x-1}p & \text{for } x = 1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

$f_X(x)$ is the size of the jump in the cdf at x. To find the probability of an event, sum the pmf over its points: for positive integers $a \le b$,

$$P(a \le X \le b) = \sum_{k=a}^{b} f_X(k) = \sum_{k=a}^{b} (1-p)^{k-1}p,$$

and as a special case,

$$P(X \le b) = \sum_{k=1}^{b} f_X(k) = F_X(b). \tag{1.6.1}$$

Convention. Uppercase letter for the cdf, the corresponding lowercase letter for the pmf or pdf: F_X and f_X.

8. Probability density functions

Point probabilities vanish for continuous random variables. Since $\{X = x\} \subset \{x - \epsilon < X \le x\}$ for any $\epsilon > 0$, Theorem 1.2.9(c) gives

$$P(X = x) \le P(x - \epsilon < X \le x) = F_X(x) - F_X(x - \epsilon)$$

for any $\epsilon > 0$. Therefore, by the continuity of F_X,

$$0 \le P(X = x) \le \lim_{\epsilon \downarrow 0} \big[F_X(x) - F_X(x - \epsilon)\big] = 0.$$

So the pmf idea fails. Instead, replace sums by integrals: just as (1.6.1) adds up point probabilities, we want

$$P(X \le x) = F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

If f_X is continuous, the Fundamental Theorem of Calculus gives

$$\frac{d}{dx} F_X(x) = f_X(x). \tag{1.6.2}$$

Definition 1.6.3. The **probability density function** or **pdf**, $f_X(x)$, of a continuous random variable X is the function that satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt \quad \text{for all } x. \tag{1.6.3}$$

A note on notation. "X has a distribution given by $F_X(x)$" is abbreviated $X \sim F_X(x)$, read "X is distributed as $F_X(x)$." Similarly $X \sim f_X(x)$, and if X and Y have the same distribution, $X \sim Y$.

A pdf is not a probability. $f_X(x)$ can exceed 1. What is true is that, for small δ, $P(x < X \le x + \delta) \approx f_X(x)\,\delta$: probability is *area* under f_X, not height.

Endpoints don't matter. Since $P(X = x) = 0$ for continuous X,

$$P(a < X < b) = P(a < X \le b) = P(a \le X < b) = P(a \le X \le b).$$

(For a discrete random variable they can all differ.)

Example 1.6.4 (Logistic probabilities). For the logistic distribution of Example 1.5.5,

$$F_X(x) = \frac{1}{1 + e^{-x}} \qquad \text{and hence} \qquad f_X(x) = \frac{d}{dx} F_X(x) = \frac{e^{-x}}{(1 + e^{-x})^2}.$$

The area under f_X gives interval probabilities (C&B Figure 1.6.1):

$$P(a < X < b) = F_X(b) - F_X(a) = \int_{-\infty}^{b} f_X(x)\,dx - \int_{-\infty}^{a} f_X(x)\,dx = \int_{a}^{b} f_X(x)\,dx.$$

The pdf (or pmf) contains the same information as the cdf, so use whichever makes the problem simpler.

9. Characterizing pmfs and pdfs

Theorem 1.6.5. A function $f_X(x)$ is a pdf (or pmf) of a random variable X if and only if

- **a.** $f_X(x) \ge 0$ for all x.
- **b.** $\sum_x f_X(x) = 1$ (pmf) or $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ (pdf).

Proof. If f_X is a pdf (or pmf), both properties follow from the definitions. For a pdf, by (1.6.3) and Theorem 1.5.3,

$$1 = \lim_{x\to\infty} F_X(x) = \int_{-\infty}^{\infty} f_X(t)\,dt.$$

Conversely, given f_X with (a) and (b), define F_X by (1.6.3) (or by summing) and check the conditions of Theorem 1.5.3. □

Normalizing. Any nonnegative function with a finite positive integral (or sum) can be turned into a pdf (or pmf). If $h(x) \geq 0$ is positive on a set A, 0 elsewhere, and

$$\int_{\{x\in A\}} h(x)\,dx = K < \infty$$

for some constant $K > 0$, then $f_X(x) = h(x)/K$ is a pdf of a random variable X taking values in A. We will use this constantly: often we know a density only "up to a constant," $f \propto h$, and K is the **normalizing constant**.

A technical caveat. Relationship (1.6.3) does not always hold: there exist continuous random variables whose cdf is continuous but not the integral of any f_X. These cases are pathological, and C&B (and we) assume that (1.6.3) holds for every continuous random variable. In more advanced texts (for example Billingsley 1995, Section 31), a random variable satisfying (1.6.3) is called **absolutely continuous**.

Check your understanding.

1. For Example 1.5.4 with $p = 0.3$, find $P(X > 3)$ in two ways: from the pmf and from the cdf.

   > From the cdf: $P(X > 3) = 1 - F_X(3) = (1-0.3)^3 = 0.343$. From the pmf: $1 - (0.3 + 0.7\times 0.3 + 0.7^2 \times 0.3) = 1 - 0.657 = 0.343$. Directly: $X > 3$ means the first three tosses are tails, probability 0.7^3.

2. Find c so that $f(x) = cx^2$ for $0 < x < 1$ (and 0 otherwise) is a pdf. Then find $F(x)$ for all x, $P(X \leq 1/2)$, and $f(0.9)$. Is $f(0.9)$ a probability?

   > $\int_0^1 cx^2\,dx = c/3 = 1$, so $c = 3$. Then $F(x) = 0$ for $x \leq 0$, $F(x) = x^3$ for $0 < x < 1$, and $F(x) = 1$ for $x \geq 1$. $P(X \leq 1/2) = 1/8$. $f(0.9) = 3 \times 0.81 = 2.43 > 1$, so it is a density, not a probability.

3. In Example 1.5.6, find $P(Y = 0)$, $P(Y < 0)$, and $P(Y \leq 0)$.

   > $P(Y < 0) = \lim_{y\uparrow 0} F_Y(y) = \frac{1-\epsilon}{2}$, $P(Y \leq 0) = F_Y(0) = \epsilon + \frac{1-\epsilon}{2} = \frac{1+\epsilon}{2}$, and $P(Y = 0) = \epsilon$, the size of the jump.

4. Which of these is a cdf? (i) $F(x) = 1 - e^{-x}$ for $x \geq 0$ and 0 for $x < 0$. (ii) $F(x) = e^{-x}$ for $x \geq 0$ and 0 for $x < 0$.

   > i. is a cdf: limits 0 and 1, increasing, continuous. It is the exponential distribution. (ii) is not: it decreases on $[0, \infty)$ and $\lim_{x \to \infty} F(x) = 0 \neq 1$.

Summary of Week 3

- **Independence:** $P(A \cap B) = P(A)P(B)$; preserved under complements; disjoint is not independent; mutual independence requires the product rule for *every* subcollection, and neither the triple product nor pairwise independence alone is enough.
- **Random variable:** a function $X : S \to \mathbb{R}$. It induces a probability function on its range by $P_X(X \in A) = P(\{s \in S : X(s) \in A\})$.
- **Cdf:** $F_X(x) = P(X \leq x)$ for all real x. A function is a cdf if and only if it has limits 0 and 1, is nondecreasing, and is right-continuous (Theorem 1.5.3). Jumps are point probabilities.
- **Continuous vs. discrete:** defined by the cdf (continuous vs. step function). The cdf determines the distribution (Theorem 1.5.10); identically distributed does not mean equal.
- **Pmf and pdf:** $f_X(x) = P(X = x)$ (discrete); $F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$ (continuous). Both are nonnegative and sum or integrate to 1 (Theorem 1.6.5). A pdf is a density, not a probability.

Practice Problems

From C&B §1.7:

- **Independence:** 1.39, 1.40, 1.44.
- **Random variables:** 1.45, 1.46.
- **Cdfs:** 1.47, 1.48, 1.49, 1.50, 1.51.
- **Pmfs and pdfs:** 1.52, 1.53, 1.54, 1.55.

Next week: distributions of functions of a random variable and expected values (§§2.1–2.2).